

The Kelly Criterion for Optimal Bet Sizing

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Consider a gamble with two states: we either win or lose.
Define the following variables:

- Let $W_0 > 0$ be your initial wealth.
- Let $x \in [0, 1]$ be the proportion of your wealth that you bet.
- Let $b > 0$ be the payout from a successful bet, so if you win you receive W_0bx and your total wealth becomes $W_0(1 + bx)$. If the odds are "x-to-y" then $b = x/y$.
- Let $a > 0$ be the proportion of your betting amount that you pay when you lose, so when you lose your wealth becomes $W_0(1 - ax)$. In an ordinary gamble $a = 1$ (you lose everything if you lose), but for a stock investment it might be reasonable to assume $a \in (0, 1)$. Also note that if you're leveraged (e.g., buying on margin), then $a > 1$.
- Let p be the probability you win and $q = 1 - p$ be the probability you lose.

Now we make three key assumptions:

1. Bets compound.
2. Infinite repeated gambles are possible.
3. Gambles are independent.

Meaning that:

1. If we win once, we have $W_0(1 + bx)$, and then $100x\%$ of our wealth becomes $W_0(1 + bx)x$, so if we lose in the next bet, we will have $W_0(1 + bx)(1 - ax)$. This assumption holds up in most real-life settings.
2. We can continue playing the gamble repeatedly forever.
3. All rounds of gambling are independent. For example, whether we win or lose the next bet is not determined by whether we won or lost any of our past bets.

Theorem (Kelly's Criterion): Under the setup above, the optimal bet size is

$$x^* = p/a - q/b$$

and in the usual case where $a = 1$ we have Kelly's formula

$$x^* = p - q/b$$

Proof: Suppose we have bet for n rounds. The number of rounds we won is a binomial random variable $N_{win} \sim \text{Bin}(n, p)$ and the number of rounds we lost is also a binomial random variable $N_{lose} = n - N_{win} \sim \text{Bin}(n, q)$. Hence, our wealth becomes

$$W_n = W_0(1 + bx)^{N_{win}}(1 - ax)^{N_{lose}}$$

so our wealth has grown by

$$R_n := \frac{W_n}{W_0} = (1 + bx)^{N_{win}}(1 - ax)^{N_{lose}}$$

which is clearly nonlinear in (N_{win}, N_{lose}) , so $\mathbb{E}(R_n)$ is difficult to calculate. Thanks to our assumptions about rounds of betting being IID, the strong law of large numbers guarantees that:

$$\frac{N_{win}}{n} \xrightarrow[n \rightarrow \infty]{a.s.} p \quad \& \quad \frac{N_{lose}}{n} \xrightarrow[n \rightarrow \infty]{a.s.} q$$

Then, the per-period (or "per-bet") growth rate is:

$$R_n^{1/n} = (1 + bx)^{N_{win}/n}(1 - ax)^{N_{lose}/n}$$

Since $R_n^{1/n}$ is a continuous function of the vector $(N_{win}/n, N_{lose}/n)$, the continuous mapping theorem guarantees that:

$$R_n^{1/n} \xrightarrow[n \rightarrow \infty]{a.s.} (1 + bx)^p(1 - ax)^q =: R$$

So our asymptotic growth rate per-round is R . We wish to find the optimal bet size x^* that maximizes R ;

$$x^* = \arg \max_x \{R\} = \arg \max_x \{\ln R\}$$

Let $L = \ln R = p \ln(1 + bx) + q \ln(1 - ax)$.

$$\frac{d}{dx} L = pb(1 + bx)^{-1} - qa(1 - ax)^{-1}$$

Then we solve for the root of the derivative:

$$\begin{aligned} \frac{d}{dx} L = 0 &\implies 0 = pb(1 + bx)^{-1} - qa(1 - ax)^{-1} \\ &\implies pb(1 - ax) = qa(1 + bx) \\ &\implies pb - pbax = qa + qabx \\ &\implies xab(q + p) = pb - qa \\ &\xrightarrow{p+q=1} xab = pb - qa \\ &\implies x = p/a - q/b \end{aligned}$$

Therefore, the optimal bet size is

$$x^* = p/a - q/b$$

This can be verified to be the maximum by calculating

$$\left. \frac{d^2 L}{dx^2} \right|_{x=x^*} = - \left(\frac{pb^2}{(1 + bx^*)^2} + \frac{qa^2}{(1 - ax^*)^2} \right) < 0$$

When $a = 1$, this simplifies to Kelly's formula $x^* = p - q/b$

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